

MADANA VENKATESH

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PROFESSIONAL SUMMARY

Third-year B.Tech student in Artificial Intelligence and Machine Learning (CGPA 8.10) with hands-on experience building and deploying AI/ML solutions and full-stack applications. Developed computer vision systems, an SVM-based healthcare predictor, and a production-grade Dynamic CMS platform during internships and freelance work. Proficient in end-to-end development using modern tools and frameworks.

SKILLS

AI & Machine Learning: Python, PyTorch (Basic), TensorFlow (Basic), Scikit-learn, Mask R-CNN, Computer Vision, Java Core

Web & Backend: FastAPI, HTML, CSS

Databases & Cloud: MongoDB, SQL, IBM Cloud, Docker

Tools & Others: CUDA, Git

EXPERIENCE

Freelance Web Developer (Part-Time)

Dec 2025 – Feb 2026

BCM Ecoweave Pvt. Ltd.

Remote

- Designed and developed a Dynamic CMS with drag-and-drop page builder supporting 23 customizable section types.
- Implemented Global Theme Engine (light/dark mode), RBAC, Media Library via Cloudinary CDN, SEO optimization, scheduling, and version history.
- Built with React 18, Vite, Tailwind CSS, Node.js, Express.js, MongoDB; secured via JWT, rate limiting; deployed on Docker, Vercel, Render.

Data Science & Analytics Intern

Aug 2025 – Nov 2025

Zidio Development

Remote

- Developed AI-based Image Captioning and Segmentation system using Transformers and Mask R-CNN; deployed via FastAPI with CUDA optimization.
- Built Healthcare Recommendation System using SVM (74% accuracy) predicting 41 diseases from 132 symptoms; integrated with Flask API.

Data Analyst Intern

Oct 2025

InnoByte Services

Remote

- Analyzed Amazon sales data using Python (Pandas, NumPy, Matplotlib, Seaborn) to identify trends in product performance and customer behavior.
- Generated insights on top-performing categories, geographical trends, and operational efficiency.

PROJECTS

Smart Parking Management System

- Built an intelligent camera-based parking system using Mask R-CNN for instance segmentation to detect vehicles and determine real-time occupancy in open parking lots.
- Implemented virtual slot mapping with IoU, vehicle size classification (Compact, Sedan, SUV), violation detection, duration tracking, and automated fee calculation.

Network Intrusion Detection System

- Developed ML-based system to detect network attacks (DoS, Probe, R2L, U2R) using Random Forest, SVM, and XG-Boost.
- Trained on KDD Cup / NSL-KDD dataset for anomaly detection and multi-class classification.

EDUCATION

Bachelor of Technology in Artificial Intelligence & Machine Learning

Dhanalakshmi Srinivasan University

Expected Graduation: 2027 — CGPA: 8.10

CERTIFICATIONS & ACHIEVEMENTS

- AWS Academy Graduate – Cloud Foundations
- MongoDB Certifications: AI Agents, Vector Search, RAG Applications, Schema Design
- Artificial Intelligence Fundamentals – IBM SkillsBuild
- Winner, Poster Presentation at International Conference (2024)
- Machine Learning for Materials Discovery Workshop (2026)

ADDITIONAL INFORMATION

Languages: English, Hindi, Telugu (Native), Tamil, Japanese (Basics)